

jcm-strat Phase 8: giving the model a QBO

Nudging the tropical stratospheric wind towards ERA5 on the Phase 6 configuration (Polvani-Kushner stratosphere, ERA5-nudged troposphere, four passive tracers, T63L95, 12 minute step). Two versions: the nudging window reaching up to 4 hPa, run for 2005 and for the 2005 to 2009 chain, and then the same runs with the window top raised to 1 hPa; then a sweep of the relaxation time (10, 5, 2, 1 days) over the same five years, after which 1 day became the default, and a mean-preserving interpolation of the monthly target. Record as of 11 September 2026, branch phase8-qbo-nudging.

1. The problem

The stripped model has no quasi-biennial oscillation. Its equatorial stratosphere sits in steady easterlies of 12 to 14 m/s with a year-to-year variability of 4 m/s where ERA5 has 17 (Phase 6 record). It cannot grow one: the QBO is driven by tropical waves launched by convection, which the dry model does not have, and by gravity-wave drag, which is off, and it needs about 500 to 700 m vertical spacing in the tropical lower stratosphere, where the L95 grid has about 1 km. For aerosol transport the QBO matters directly. It modulates the tropical upwelling and the subtropical mixing barrier, so an injected layer's residence time in the tropics and its leak to mid-latitudes differ by tens of percent between the easterly and the westerly phase; Pinatubo showed it and every injection study reports it. Under specified dynamics the standard remedy is to relax the tropical wind towards the observed one, as WACCM does in its QBO-nudged configurations. That is what Phase 8 adds, and it measures what changes.

2. Everything the model is relaxed towards, in one table

A relaxation adds a tendency $-(x - \text{target})/\tau$ to a field x : the field is pulled towards the target with e-folding time τ . The stripped model is held together by the six relaxations in the table, acting on different fields in different layers. The QBO nudging of this phase is one of them; the others are unchanged from Phase 6. 'Target cadence' says how often the target itself carries new information; between samples every target is interpolated linearly in time to the 12 minute step.

field	where	target	tau	target cadence
u, v, T (full fields)	troposphere: pressure above 150 hPa, except the two lowest model levels	ERA5 (WeatherBench2 6-hourly, regridded to the model grid and levels)	6 h	6-hourly samples
T (full field)	stratosphere: pressure below 100 hPa, all latitudes	Polvani-Kushner (2002) seasonal equilibrium: US standard atmosphere outside the winter polar cap, a vortex profile cooling at 4 K/km inside it (cap edge 50 deg, width 10 deg), the cooling faded out above 3 hPa	15 d	analytic; follows the calendar
T (full field)	troposphere, underneath the ERA5 nudging	Held-Suarez equilibrium, floored by the standard atmosphere	40 d in the free troposphere, 4 d in the boundary layer (sigma above 0.7)	analytic (the 6 h nudging dominates)
u, v (Rayleigh friction)	boundary layer, sigma above 0.7	zero	1 d at the surface, weakening linearly to none at sigma 0.7	-
zonal-mean u only	tropics: full weight within 15 deg of the equator, cos-squared taper to zero at 25 deg; pressure 90 hPa up to the window top (4 hPa in the first version, 1 hPa in the second), full weight from 40 hPa to about 9 hPa (2.2 hPa in the second version), log-linear tapers	ERA5 monthly-mean zonal-mean zonal wind (CDS pressure-level product, 25 levels 1000 to 1 hPa, 2005 to 2009)	10 d in the first version; 1 d since section 7	monthly means, interpolated between month centres (mean-preserving nodes since section 8)

field	where	target	tau	target cadence
u, v to zero; T to its zonal mean and to 250 K (sponge)	the top 10 model levels, 0.01 to 0.15 hPa (above about 65 km)	zero wind; zonal-mean T and 250 K	1.5 h at the top level, doubling with each level downward (32 d at the tenth)	-

In altitude (7 km scale height): 150 hPa is about 13 km, 90 hPa 17 km, 40 hPa 22.5 km, 9 hPa 33 km, 4 hPa 39 km, 2.2 hPa 43 km, 1 hPa 48 km. The model lid is 0.01 hPa, about 80 km. In its first version the QBO nudging was the slowest data-driven relaxation in the model (10 days against 6 hours in the troposphere) and the one with the coarsest target (monthly against 6-hourly); sections 7 and 8 change the first and keep the second on purpose.

3. The recipe, and why each choice

- **Target: ERA5 monthly zonal means from the Copernicus Climate Data Store.** ERA5's tropical stratospheric winds are observation-constrained (the Singapore and tropical radiosonde network, GNSS radio occultation) and its QBO is right. The CDS monthly product reaches 1 hPa, whereas the WeatherBench2 store that feeds the tropospheric nudging has nothing usable above about 60 hPa, which is why the QBO could not simply be included in that nudging. ERA5 is also what the troposphere follows, so one reanalysis constrains the whole column, and the years line up with the CLaMS age-of-air reference. A WACCM target would have imposed a modelled, weaker QBO; a climatological or idealised one would have lost the real 2005 to 2009 phase sequence.
- **What is relaxed: only the zonal mean of the zonal wind.** At every longitude of a latitude row the same tendency, $-(u_{\text{bar_model}} - u_{\text{bar_ERA5}})/\tau$, is applied. The waves that do the transport are left alone; only the mean flow they propagate through is corrected. Relaxing the full wind field would damp the eddies too.
- **Where: the QBO's own domain.** Full weight within 15 degrees of the equator, fading to zero at 25; full weight between 40 and 9 hPa, fading to zero at 90 hPa (clear of the 150 hPa tropospheric nudging) and at 4 hPa (below the semiannual oscillation region). Section 6 raises the top to 1 hPa.
- **How fast: tau = 10 days,** WACCM's choice. Slow enough not to fight the resolved waves step by step, fast enough to hold the observed phase. It also means the model is *guided*, not prescribed: its own dynamics can and do pull back, which is why the amplitude comes out at 80 percent of the target (section 5) and why a shorter tau is the obvious next knob (section 7).
- **Implementation.** `jcm_strat/qbo_nudging.py`; the term runs on the column path, the zonal mean is a segment mean over the columns of each latitude row, the fraction of year comes from the same clock the seasonal Polvani-Kushner term uses, and the calendar year is a configuration argument that the segment chain passes per year. The QBO tendency is computed inside the Polvani-Kushner term (PolvaniKushnerQbo, one term instead of two); three CPU tests check the weights, that the tendency equals $-k w (u_{\text{bar}} - \text{target})$ exactly, and that the combined term equals the two-term sum. Configuration: `physics/strat_pk_qbo.yaml`, experiment `p8_qbo`.
- **One hazard.** The first three attempts died with RESOURCE_EXHAUSTED allocating 29 GiB. JAX preallocates 75 percent of the card by default; the compiled step holds the 29 GB one-year nudging target and at times a second copy of it, so the Phase 6 configuration sat just below that ceiling and any additional term tipped it over. `scripts/env.sh` now sets `XLA_PYTHON_CLIENT_MEM_FRACTION=0.92`. Nothing about the physics changed.

4. The runs

run	window top	length	purpose	compared with
p8_qbo_2005_oom, _oom2	4 hPa	0	GPU out of memory under the default 75 percent preallocation	-
p8_qbo_2005	4 hPa	1 year (2005)	first year with QBO nudging	p6_pk_g4_t15_s05_top3 (Phase 6, same year)
p8_2005 to p8_2009, aggregate p8_5yr	4 hPa	5 years	the chain with tracers (chain_years.sh, 82 min on GPU 0)	p6_5yr (Phase 6 chain)

run	window top	length	purpose	compared with
p8b_2005 to p8b_2009, aggregate p8b_5yr	1 hPa	5 years	the same chain with the window top raised to 1 hPa (section 6)	p8_5yr and p8_qbo_2005

All runs: T63L95, 12 minute step, ERA5-nudged troposphere below 150 hPa, Polvani-Kushner stratosphere as chosen in Phase 6, four passive tracers, GPU 0. The chains are five one-year segments, each restarted from the previous checkpoint.

5. Results with the window top at 4 hPa

Acceptance checks proposed before the runs, all on the 2005 to 2009 chain unless stated:

check	threshold	result
equatorial (5S-5N) monthly u vs ERA5, 10-70 hPa	RMS below 5 m/s (Phase 6: 16)	pass : 3.9 m/s (2005 alone 4.2; Phase 6 chain 16.3)
QBO amplitude: deseasonalised std at 20 and 30 hPa	within 30 percent of ERA5	pass : 14.1 / 12.4 m/s vs ERA5 17.5 / 15.2 (81, 82 percent); 10 hPa 78 percent, 50 hPa 73 percent. Phase 6: 3.9 / 3.7
the change is confined to the tropics	RMS change in time-mean u outside 30 deg, 1-100 hPa, below 2 m/s; troposphere unchanged	pass : 0.5 m/s outside, 0.0 in the troposphere, 3.3 inside the window
polar vortex, sudden warmings	unchanged within noise	pass : DJF u(60N, 10 hPa) 32 m/s (Phase 6: 32, ERA5 28); the January 2009 warming moves 5 days closer to ERA5, a spurious March 2005 reversal disappears; 2006 and 2007 still missed (issue 33)
stratospheric climatology vs ERA5, 100-1 hPa	not worse than Phase 6	pass on T : 6.3 K (6.4); u 6.4 m/s (5.6) : the extra error is the tropical upper stratosphere, see below
tracer conservation	Phase 3 levels	pass : unity max deviation 2.6e-4, sai burden -0.8 percent vs analytic, minima at or above 0, top-level polar sai 8 percent of the column
age of air	reported	tropics 2.16 yr at 55 hPa (Phase 6: 2.29; CLaMS 1.33); contrast to 50-70 deg 1.63 (1.56; 2.79); at 12 hPa 3.61 (3.70; 3.68)
throughput	within 5 percent of Phase 6	fail, -13 percent : 3880 vs 4445 days/hr stepping (8 vs 7 ms per step); the zonal-mean reduction costs about 1 ms, issue 44

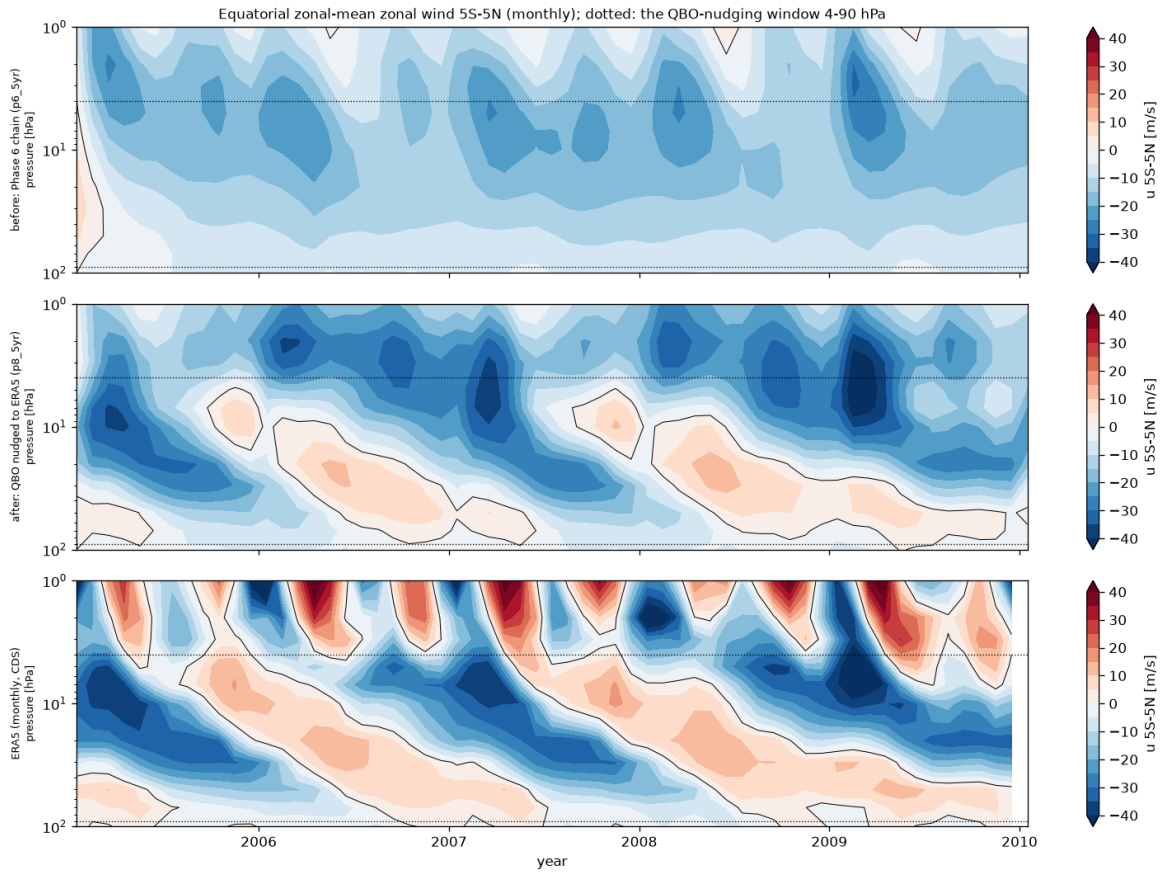


Figure 1. Equatorial (5S to 5N) zonal-mean zonal wind, monthly, 2005 to 2009, 100 to 1 hPa. Top: the Phase 6 chain, steady easterlies. Middle: the QBO-nudged chain. Bottom: ERA5. Dotted lines mark the nudging window (90 to 4 hPa). The descending easterly and westerly shear zones are reproduced in phase. Two shortfalls are visible: the westerly phases are weaker than ERA5's (+5 to +10 m/s where ERA5 has +10 to +20), and above the window the model is more easterly than ERA5 and misses the semiannual oscillation at 1 to 3 hPa entirely.

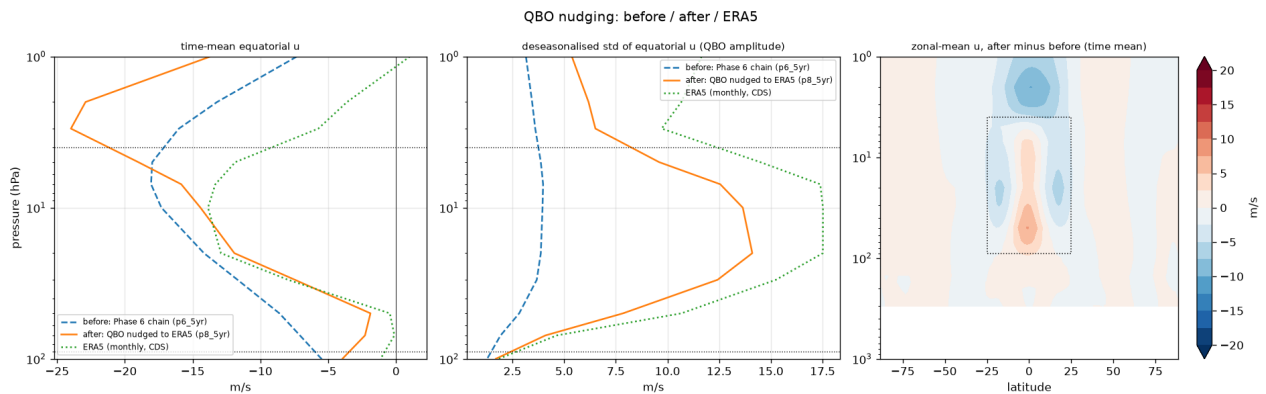


Figure 2. Left: time-mean equatorial wind. Middle: its deseasonalised standard deviation, the QBO amplitude, 80 percent of ERA5's inside the window. Right: the change in time-mean zonal-mean wind, nudged minus Phase 6, with the window dotted: the change is confined to the tropics, apart from the easterly anomaly above the window at 1 to 3 hPa (10 to 15 m/s more easterly than before; -24 m/s at 3 hPa against ERA5's -8). Momentum is carried upward out of the nudged layer and nothing above opposes it, which is the whole of the 0.8 m/s global wind-error increase in the acceptance table.

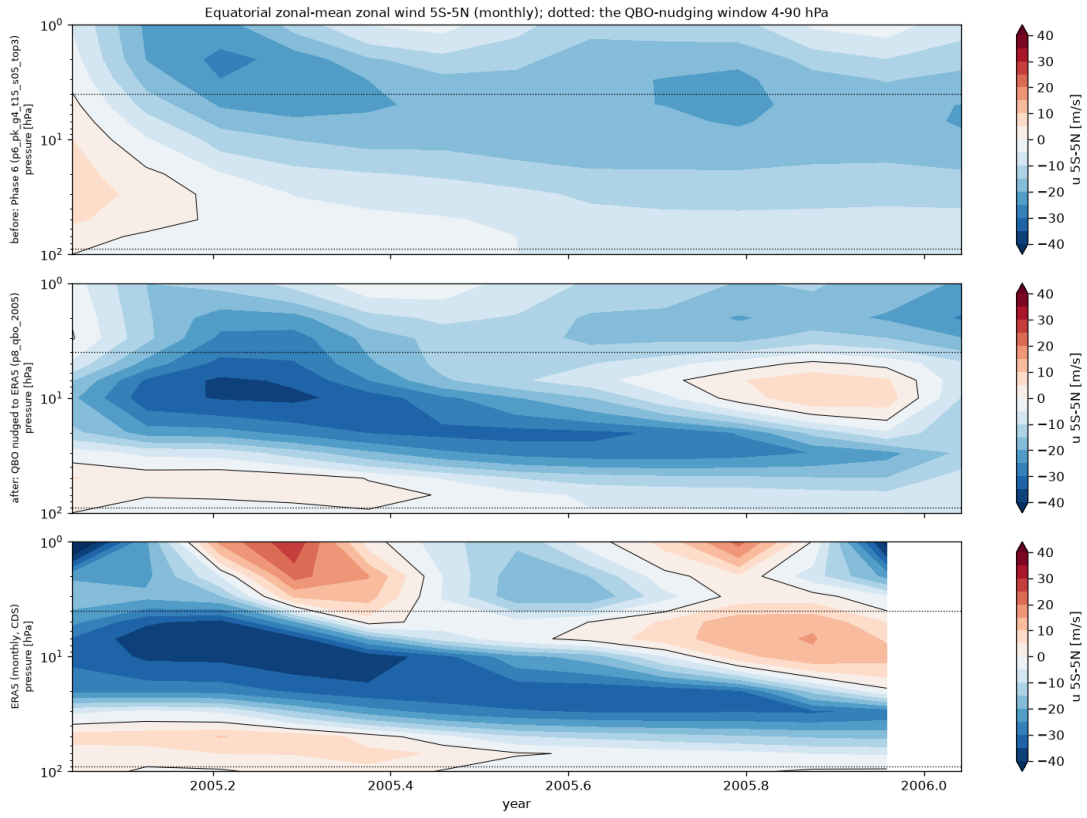


Figure 3. The single year 2005 (Phase 6 run, nudged run, ERA5). After 30 days the wind at 10 / 20 / 30 hPa reads -29 / -21 / -6 m/s against ERA5's -31 / -26 / -6 (Phase 6: +2 / +5 / +6). The nudging locks on within a month.

What the rest of the stratosphere and the tracers did

The five-year mean transport moves a little, in the right direction, and not much: the QBO mainly redistributes transport between its phases, and a five-year mean averages over two cycles. Tropical age of air at 20 km drops from 2.29 to 2.16 years, the tropics-to-extratropics contrast rises from 1.56 to 1.63 years (CLaMS 2.79), and the Brewer-Dobson upwelling at 70 hPa strengthens by 7 percent (7.2 to 7.7×10^9 kg/s; WACCM6 6.1). The 0.9 year tropical excess that remains is the Phase 4 diagnosis unchanged: slow transit through the unmixed troposphere and lower stratosphere (issue 25), not the tropical wind. The QBO's value for aerosol is in phase-dependent statements (residence time in the easterly against the westerly phase, subtropical leakage), which this configuration can now make and the Phase 6 one could not.

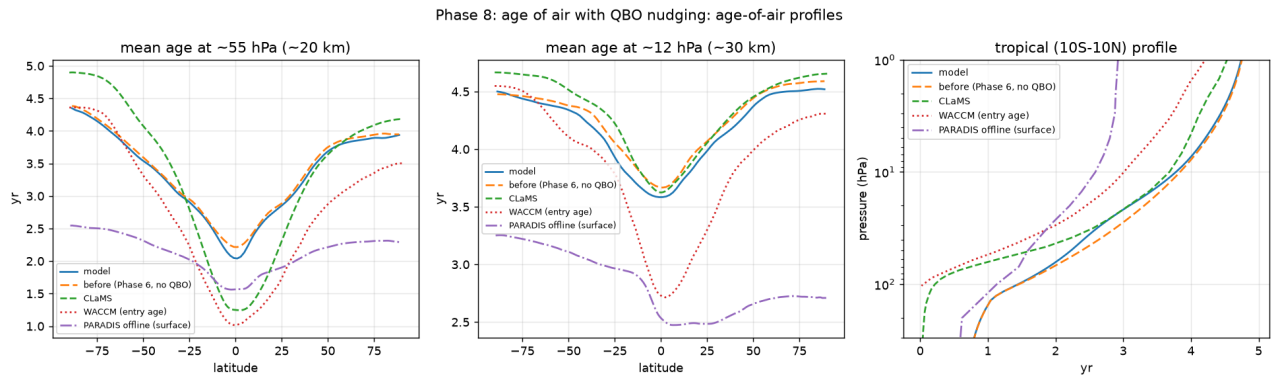


Figure 4. Mean age of air at about 20 km (left) and 30 km (middle), and the tropical profile (right): nudged chain (blue), Phase 6 chain (orange dashed), CLaMS (green), WACCM6 entry age (red), and the offline clock carried by PARADIS winds (violet). The QBO nudging changes the age by 0.1 year in the tropics and nothing elsewhere.

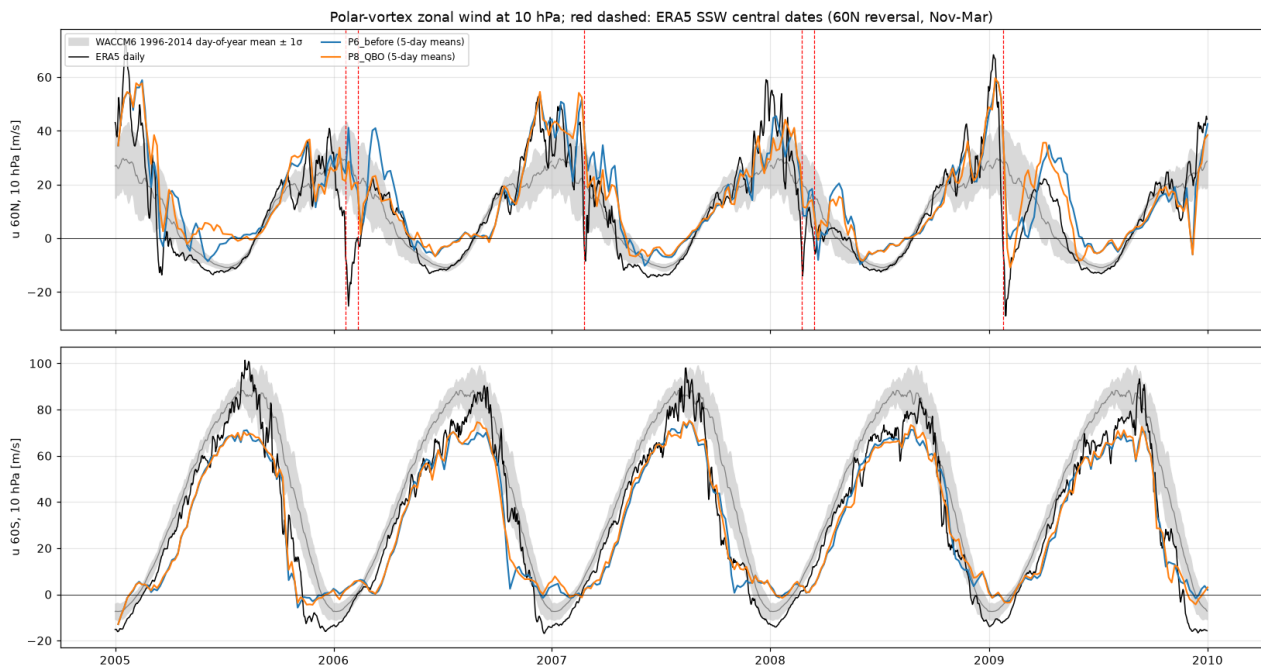


Figure 5. Polar-vortex wind at 60N (top) and 60S (bottom), 10 hPa: nudged chain (orange), Phase 6 chain (blue), ERA5 (black), WACCM6 envelope (grey), ERA5 major warmings dashed red. The two model curves lie on top of each other outside the tropics: the nudging does what it says, the tropical mean flow and nothing else.

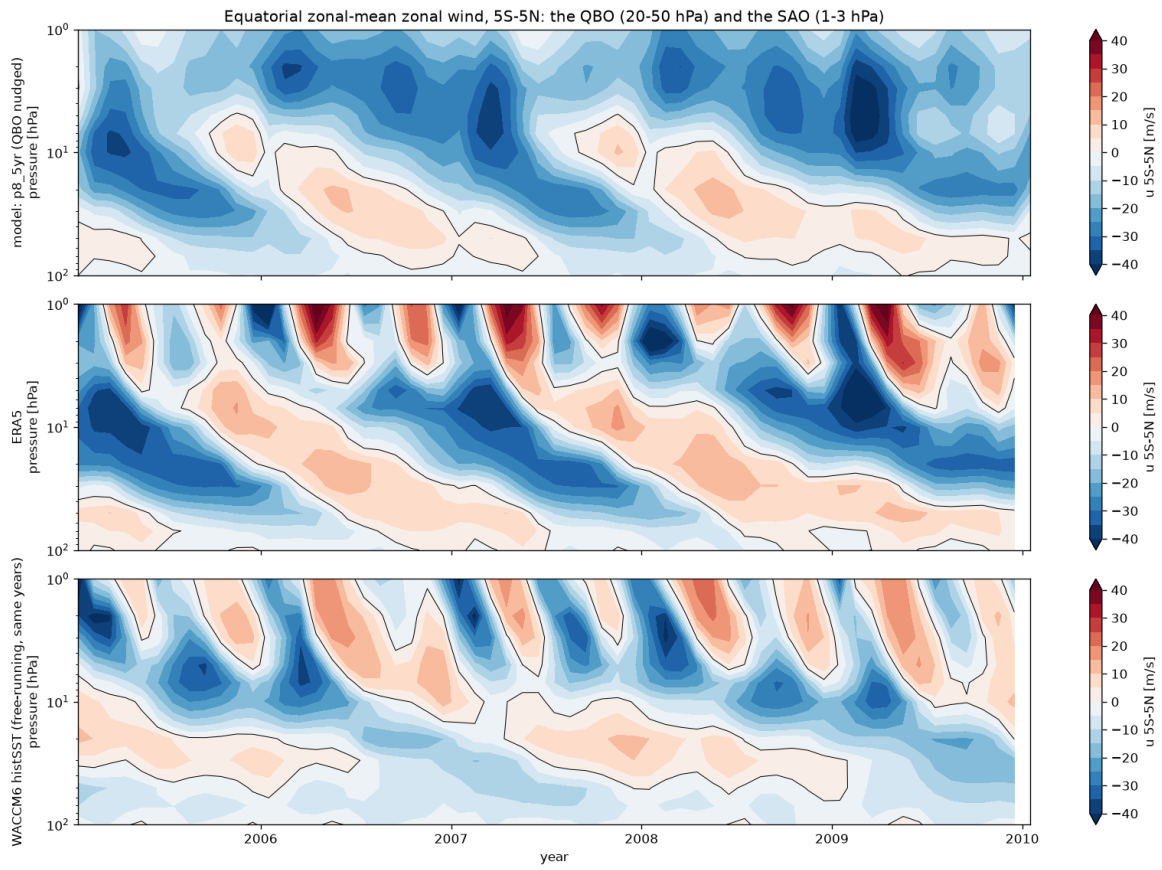


Figure 6. The same equatorial section as Figure 1 with WACCM6 histSST (free-running, same years) as the third panel instead of the Phase 6 chain. WACCM's own QBO is weaker than ERA5's and out of phase with the real one, which is why WACCM was not used as the target; its semiannual oscillation at 1 to 3 hPa, however, is present, and ours is not.

6. Raising the window top to 1 hPa

Two defects of the first version point at the top of the window. The westerly phase of the QBO first appears at 5 to 10 hPa and descends from there; with the taper ending at 4 hPa that onset layer was only partly nudged (weight 0.3 at 5 hPa, 0.7 at 7 hPa). And above the window the mean flow had become 10 to 15 m/s more easterly than ERA5 (Figure 2), a reservoir of easterly momentum sitting directly on top of the onset layer. Both argue for extending the window upward. The ERA5 monthly target already reaches 1 hPa and the model lid is 0.01 hPa, so this is one configuration key: `p_top_hpa` 4 becomes 1. With the existing taper the weight is then 1 from 40 hPa up to 2.2 hPa and falls to zero at 1 hPa; the semiannual oscillation at 1 to 3 hPa, whose monthly means ERA5 resolves, is nudged in with it. Nothing else changed: same tau, same latitude window, same target, same years. Runs `p8b_2005` to `p8b_2009`, aggregate `p8b_5yr`, compared against the 4 hPa runs. One segment died once in JCM's provenance probe, which decodes the working-tree git diff as UTF-8 and met a modified tracked PDF; `.gitattributes` now marks PDFs binary and the segment was rerun.

The single year first

2005	std 10 / 20 / 30 / 50 hPa [m/s]	mean u 20 / 30 hPa	RMS vs ERA5, 10-70 hPa	RMS vs ERA5, 1-7 hPa
window top 4 hPa (<code>p8_qbo_2005</code>)	15.5 / 8.2 / 8.8 / 7.8	-23.0 / -18.7	4.2	17.5
window top 1 hPa (<code>p8b_2005</code>)	15.8 / 8.2 / 8.8 / 7.8	-23.1 / -18.7	4.2	10.6
ERA5 (monthly, CDS)	19.7 / 9.7 / 10.5 / 10.3	-27.3 / -22.0	-	-

Single year, so the standard deviation is the plain one, not deseasonalised. RMS change in the time-mean zonal-mean wind, 1 hPa minus 4 hPa version: 2.7 m/s inside the window, 0.5 outside it in the stratosphere, 0.0 in the troposphere.

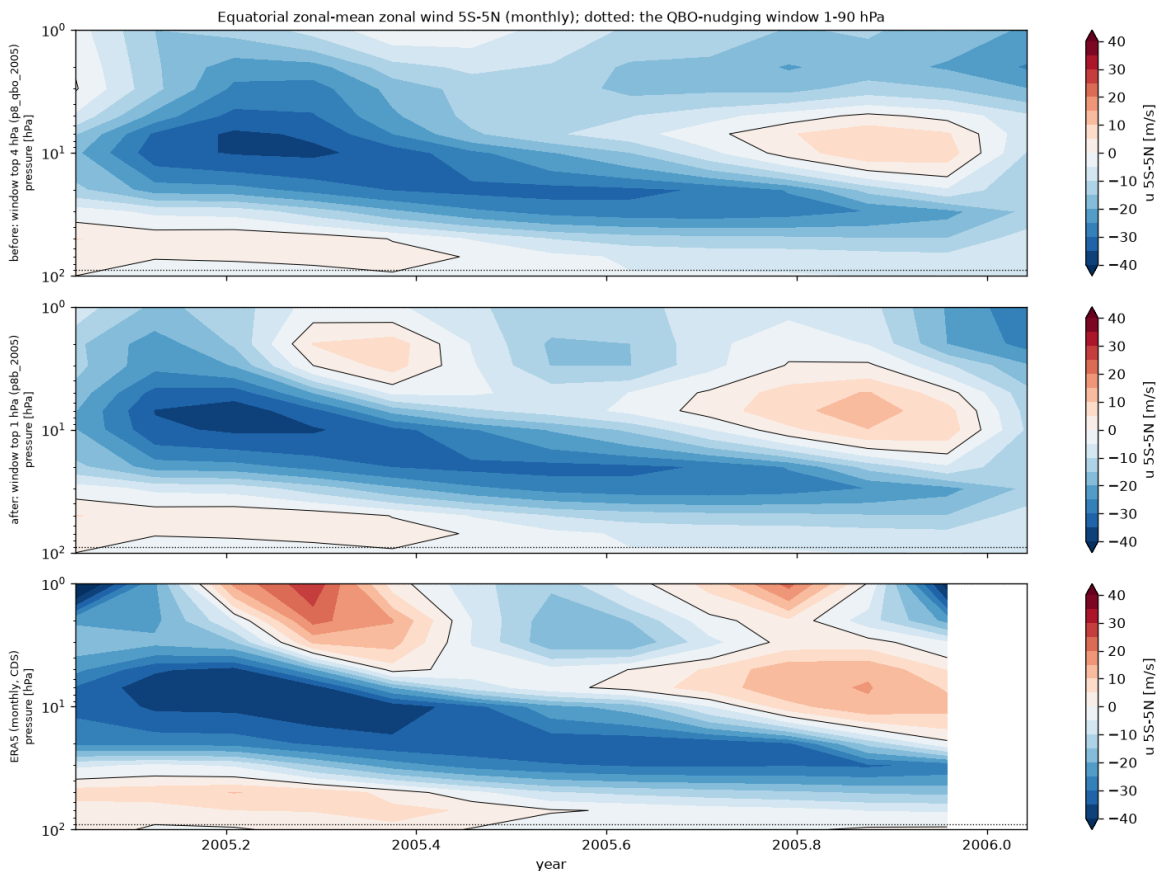


Figure 7. 2005 with the window top at 4 hPa (top), at 1 hPa (middle), and ERA5 (bottom). Inside the QBO layer the two runs are the same (RMS 4.2 m/s in both). Above 5 hPa the 1 hPa version follows ERA5: the error over 1 to 7 hPa falls from 17.5 to 10.6 m/s, the westerly onset of late 2005 at 5 to 10 hPa is stronger (+10 m/s where the 4 hPa run had +5), and the April semiannual westerlies at 1 to 2 hPa appear, at a fifth of ERA5's amplitude. The remaining upper-level error is what a 10-day relaxation leaves of a semiannual signal that the model's own dynamics pull against.

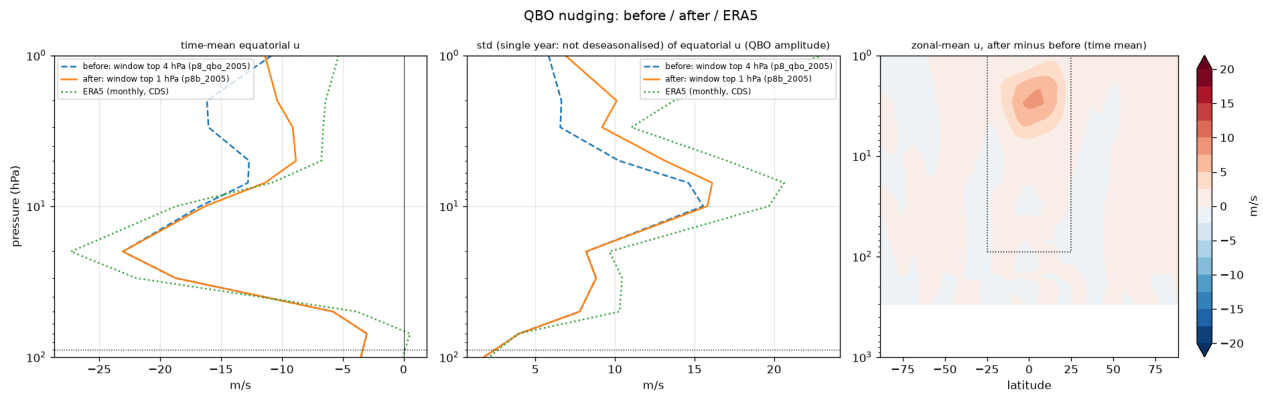


Figure 8. 2005 profiles. The time-mean wind at 1 to 3 hPa moves from -16 m/s to -10 (ERA5 -6); the variability above 5 hPa doubles; the change (right) is a single patch centred at 2 to 3 hPa, +10 m/s, and nothing elsewhere.

The five-year chain

2005-2009	deseasonalised std 10 / 20 / 30 / 50 hPa [m/s]	mean u 20 / 30 hPa	RMS vs ERA5, 10-70 hPa	RMS vs ERA5, 1-7 hPa	SAO amplitude 1 / 2 / 3 hPa
window top 4 hPa (p8_5yr)	13.6 / 14.1 / 12.4 / 7.8	-11.9 / -7.5	3.9	22.7	4.7 / 5.6 / 5.5
window top 1 hPa (p8b_5yr)	13.7 / 14.1 / 12.4 / 7.8	-11.9 / -7.5	4.0	12.5	9.5 / 15.8 / 13.8
ERA5 (monthly, CDS)	17.5 / 17.5 / 15.2 / 10.7	-12.9 / -8.0	-	-	30.7 / 20.7 / 15.6

Time-mean equatorial wind at 3 hPa: -24 m/s with the 4 hPa top, -9 with the 1 hPa top, ERA5 -6. RMS change in the time-mean zonal-mean wind, 1 hPa minus 4 hPa version: 5.0 m/s inside the window (one patch at 2 to 3 hPa), 0.5 outside it in the stratosphere, 0.0 in the troposphere. SAO: amplitude of the semiannual harmonic of the equatorial wind (strat_circulation.py).

check	window top 4 hPa	window top 1 hPa	reference
stratospheric climatology vs ERA5, 100-1 hPa, annual	T 6.3 K, u 6.4 m/s	T 6.3 K, u 4.9 m/s	Phase 6: 6.4 K, 5.6 m/s
polar vortex, DJF u(60N, 10 hPa) / JJA u(60S, 10 hPa)	32 / 65 m/s	31 / 64 m/s	ERA5 28 / 72
SSW-like reversals	2008-03-21, 2009-01-31, 2009-12-07	2008-03-26, 2009-01-31, 2009-12-07	ERA5 majors 2006-01, 2007-02, 2008-02, 2009-01
Brewer-Dobson upward mass flux at 70 hPa, DJF / JJA / annual	9.2 / 6.7 / 7.7	9.4 / 6.6 / 7.7	WACCM6 8.6 / 5.4 / 6.1 ($\times 10^9$ kg/s)
age of air at 20 km, tropics / 50-70 deg / contrast	2.16 / 3.79 / 1.63 yr	2.16 / 3.81 / 1.64 yr	CLaMS 1.33 / 4.12 / 2.79
age of air at 30 km, tropics	3.61 yr	3.68 yr	CLaMS 3.68
tracer conservation: unity max deviation, sai vs analytic, polar top-level sai	2.6e-4, -0.8 percent, 8 percent	2.7e-4, -0.8 percent, 7.6 percent	Phase 3 levels
throughput, stepping / end-to-end	3820-3960 / 2120-2190 days per hour	3890-3930 / 1870-1900 days per hour	Phase 6: 4445 / 2090

End-to-end throughput of the 1 hPa chain is lower only because of compile and output time in a busier session; the stepping rate, 8 ms per step, is the same.

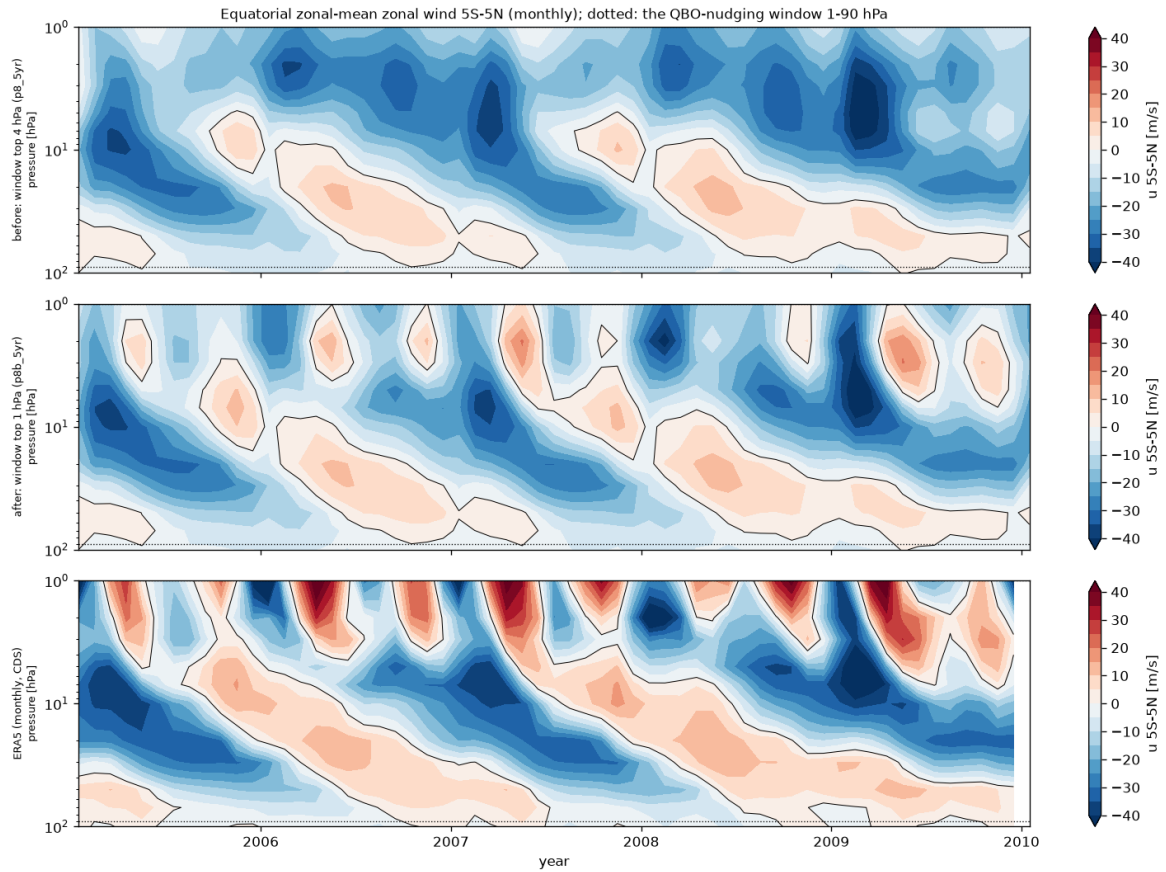


Figure 9. Equatorial wind 2005 to 2009: window top 4 hPa (top), 1 hPa (middle), ERA5 (bottom); the dotted lines are now 90 and 1 hPa. Below 5 hPa the two model panels are the same picture. Above it the 1 hPa version follows ERA5: the easterly reservoir at 1 to 4 hPa is gone and the alternating semiannual westerlies and easterlies at 1 to 3 hPa appear, at roughly three quarters of ERA5's amplitude at 2 to 3 hPa and a third at 1 hPa, where the taper reaches zero.

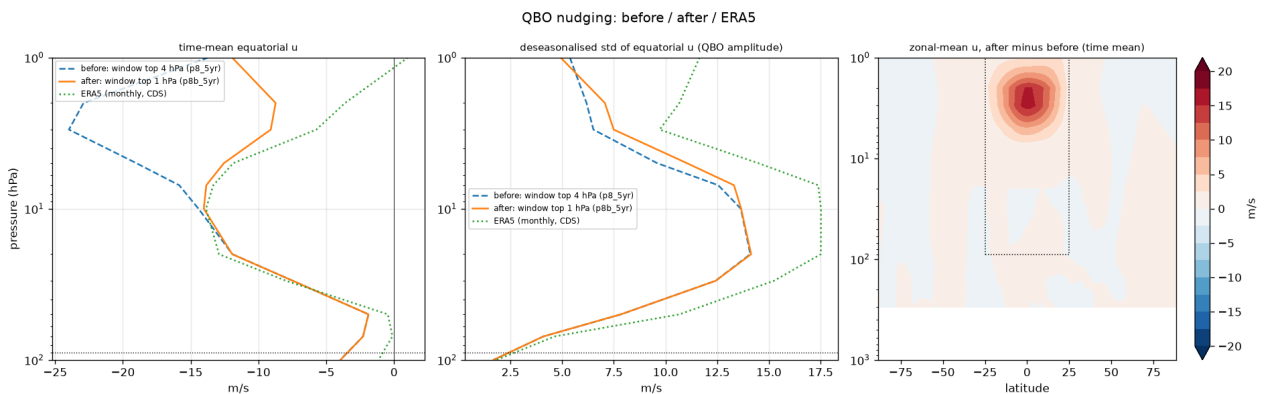


Figure 10. Left: time-mean equatorial wind; the 1 hPa version (orange) sits on ERA5 (green) from 40 hPa to 2 hPa, the 4 hPa version (blue dashed) departs above 5 hPa. Middle: QBO amplitude, identical below 5 hPa, doubled above. Right: the change in time-mean zonal-mean wind is one patch at 2 to 3 hPa over the equator, +15 m/s, and nothing anywhere else.

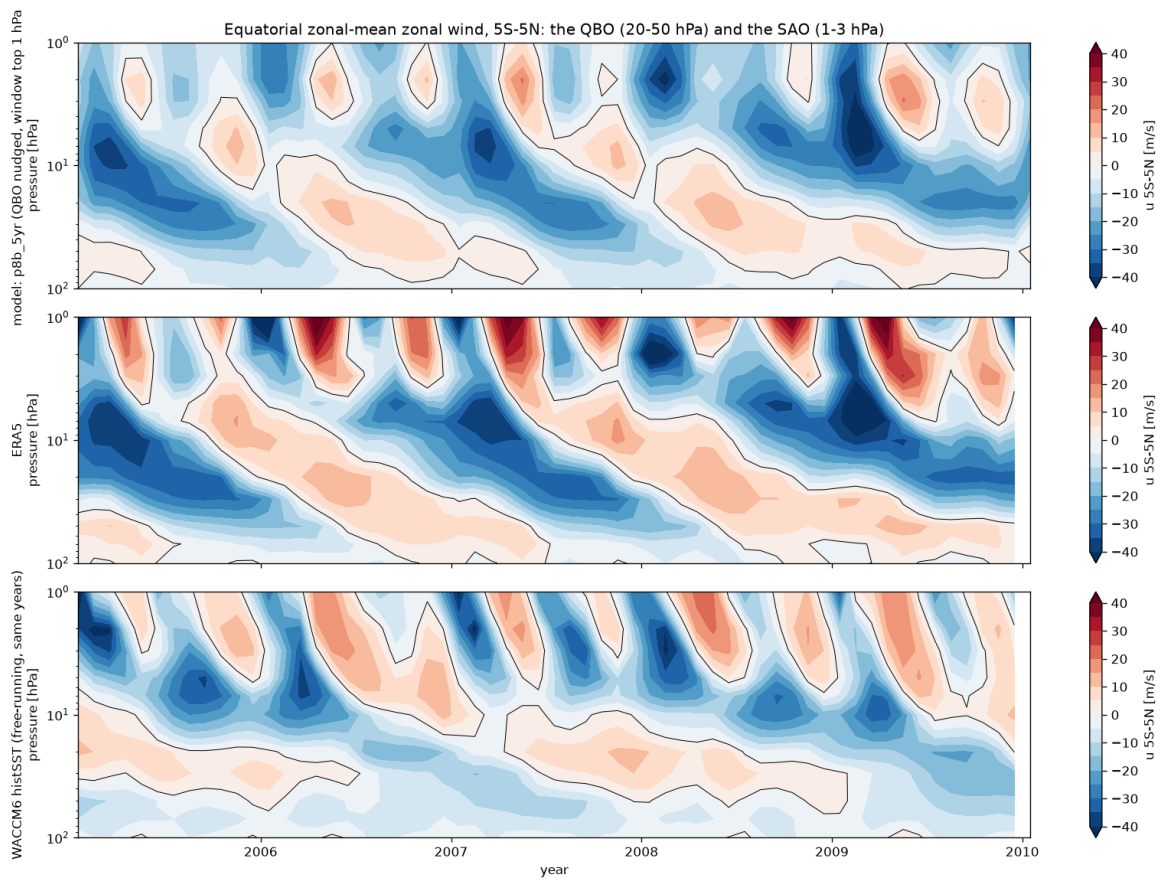


Figure 11. The 1 hPa chain against ERA5 and WACCM6. Compare with Figure 6: the semiannual oscillation that was missing is now present; it is weaker than WACCM's at 1 hPa and comparable at 2 to 3 hPa.

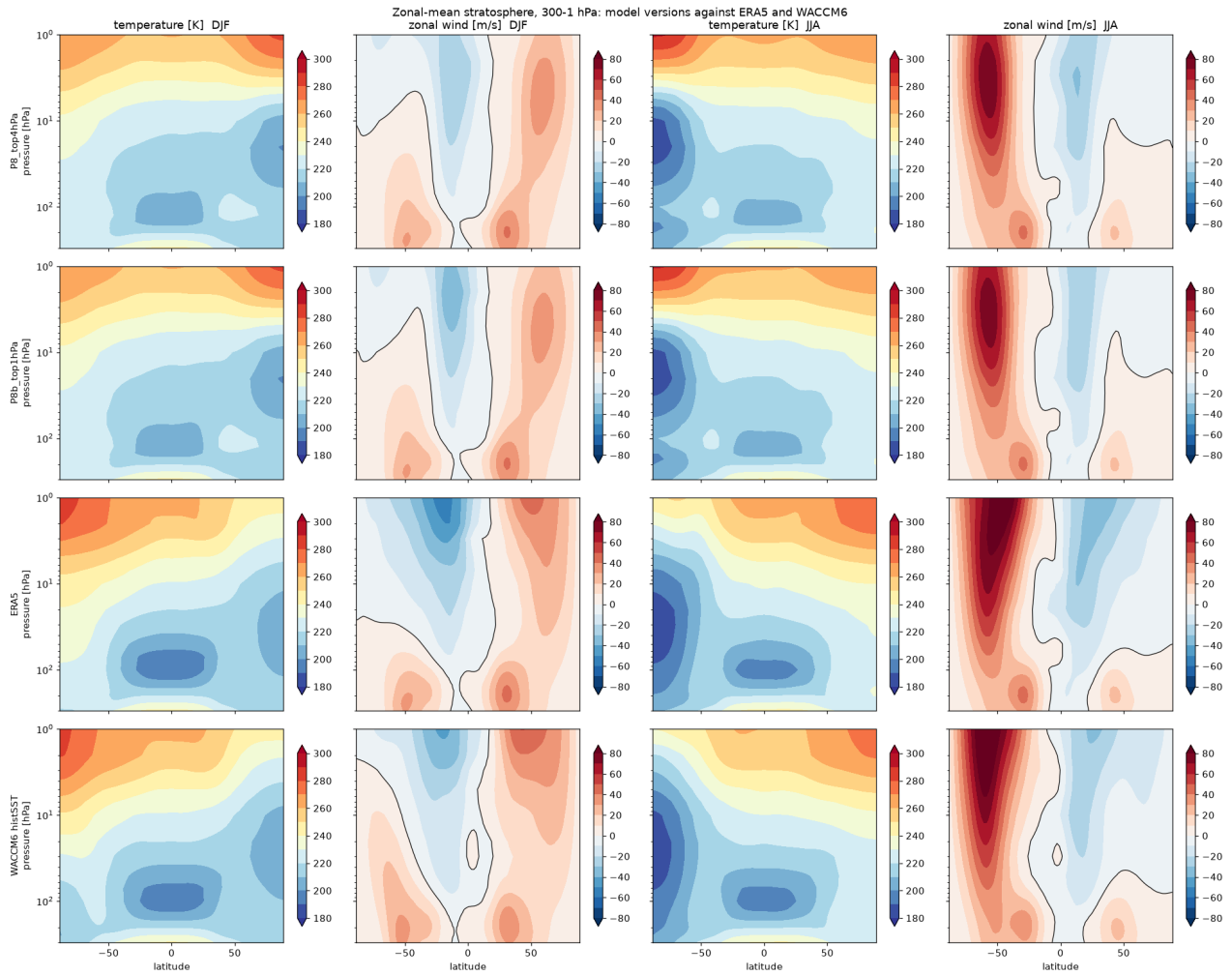


Figure 12. Zonal-mean temperature and wind, DJF and JJA, 300 to 1 hPa: 4 hPa version, 1 hPa version, ERA5, WACCM6. The only visible difference between the two model rows is the equatorial wind above 5 hPa in DJF, where the 1 hPa version has ERA5's weak westerlies instead of easterlies. Everything poleward of 25 degrees is identical.

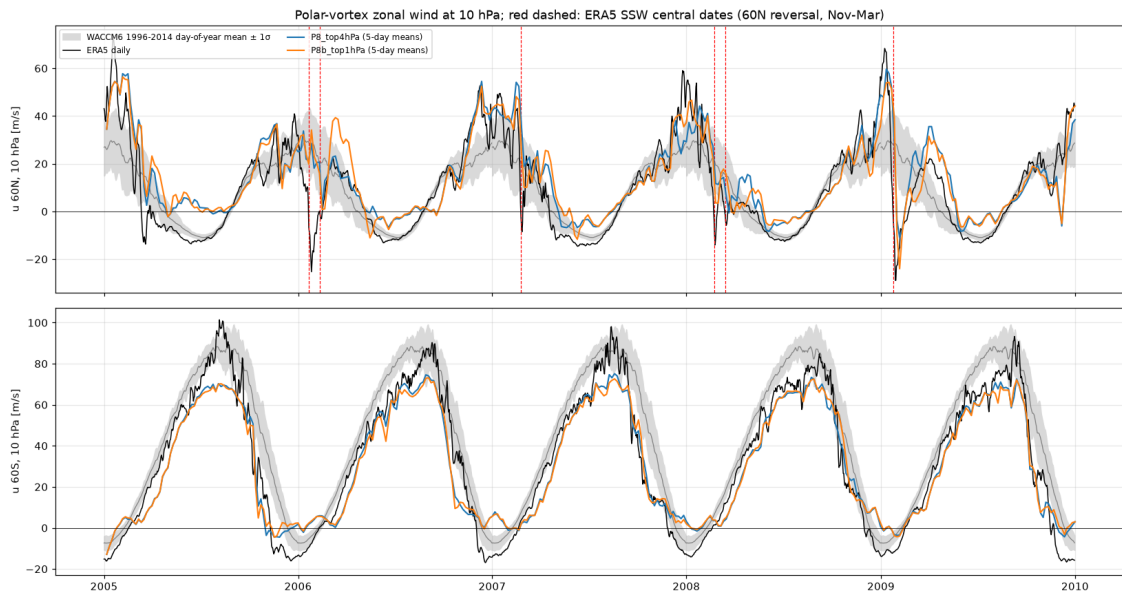


Figure 13. Polar-vortex wind at 10 hPa, 60N and 60S, both versions against ERA5 and WACCM6. The two curves lie on top of each other.

Phase 8: QBO nudging, window top 1 hPa: age-of-air profiles

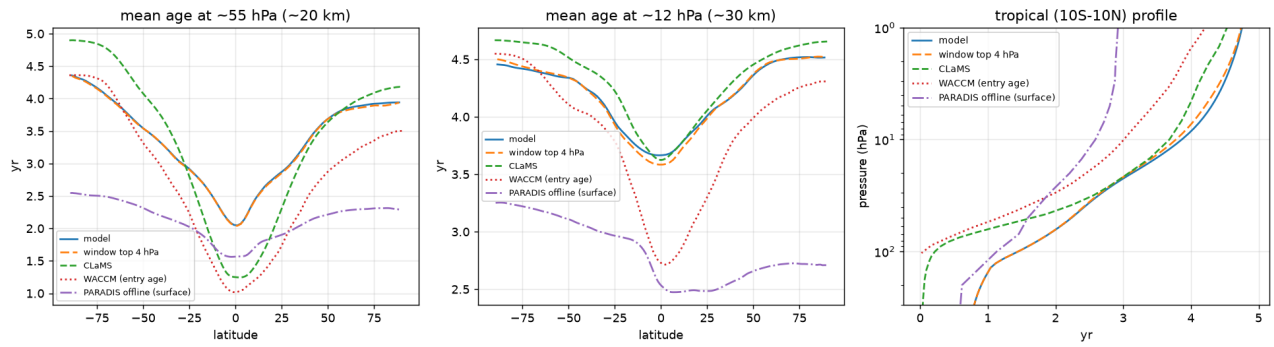


Figure 14. Age of air, 1 hPa version (blue) against the 4 hPa version (orange dashed), CLaMS, WACCM6 and the PARADIS offline clock. Indistinguishable: the tropical age at 20 km is 2.16 years in both.

7. A shorter relaxation time: the tau sweep, 10 to 1 days

Raising the window top left the QBO layer exactly as it was, so the 80 percent amplitude is the relaxation, not the geometry. Two springs act on the tropical zonal-mean wind: the nudging, pulling toward ERA5 with time constant τ , and the model's own dynamics, pulling toward the model's own easterly state (tropical upwelling lifting slow-spinning air, and the 15-day Polvani-Kushner temperature relaxation eroding the warm and cold layers a westerly jet needs). The wind settles at the weighted mean, so the swings shrink by the factor $\tau_{\text{model}} / (\tau + \tau_{\text{model}})$. Eighty percent at τ 10 days implies a model restoring time of about 40 days and predicts 89 percent at 5 days, 95 at 2 days, 99 at 6 hours. The monthly target is kept: the QBO descends about 1 km a month, so a monthly target interpolated between month centres resolves its transitions to within a week, and a daily one would only add sub-monthly wind changes that are not QBO. One key: τ_{days} 5. Runs p8c_2005 to p8c_2009 (aggregate p8c_5yr) on GPU 2 while Phase 9 occupied GPU 0; before-state p8b_5yr (τ 10 days, window top 1 hPa).

2005-2009	deseasonalised std 10 / 20 / 30 / 50 hPa [m/s]	RMS vs ERA5, 10-70 hPa	RMS vs ERA5, 1-7 hPa	SAO amplitude 1 / 2 / 3 hPa
τ 10 d (p8b_5yr)	13.7 / 14.1 / 12.4 / 7.8	4.0	12.5	9.5 / 15.8 / 13.8
τ 5 d (p8c_5yr)	14.6 / 15.1 / 13.2 / 8.5	3.1	10.9	12.2 / 17.3 / 13.9
ERA5 (monthly, CDS)	17.5 / 17.5 / 15.2 / 10.7	-	-	30.7 / 20.7 / 15.6

Amplitude at 20 hPa as a fraction of ERA5: 81 to 86 percent (prediction 89). Change in the time-mean zonal-mean wind, τ 5 minus τ 10: 0.7 m/s inside the window, 0.2 outside it in the stratosphere, 0.0 in the troposphere.

check	τ 10 d	τ 5 d	reference
stratospheric climatology vs ERA5, 100-1 hPa, annual	T 6.3 K, u 4.9 m/s	T 6.2 K, u 4.8 m/s	Phase 6: 6.4 K, 5.6 m/s
polar vortex, DJF u(60N, 10 hPa) / JJA u(60S, 10 hPa)	31 / 64 m/s	30 / 64 m/s	ERA5 28 / 72
SSW-like reversals	2008-03-26, 2009-01-31, 2009-12-07	2006-02-15, 2008-03-21, 2009-01-31, 2009-12-07	ERA5 majors 2006-01/02, 2007-02, 2008-02, 2009-01
Brewer-Dobson upward mass flux at 70 hPa, DJF / JJA / annual	9.4 / 6.6 / 7.7	9.4 / 6.8 / 7.9	WACCM6 8.6 / 5.4 / 6.1 ($\times 10^9$ kg/s)
age of air at 20 km, tropics / 50-70 deg / contrast	2.16 / 3.81 / 1.64 yr	2.15 / 3.80 / 1.65 yr	CLaMS 1.33 / 4.12 / 2.79
tracer conservation: unity max deviation, sai vs analytic, polar top-level sai	2.7e-4, -0.8 percent, 7.6 percent	2.8e-4, -0.8 percent, 7.3 percent	Phase 3 levels
throughput, stepping	3890-3930 days per hour, 8 ms per step	3910-3940 days per hour, 8 ms per step	Phase 6: 4445

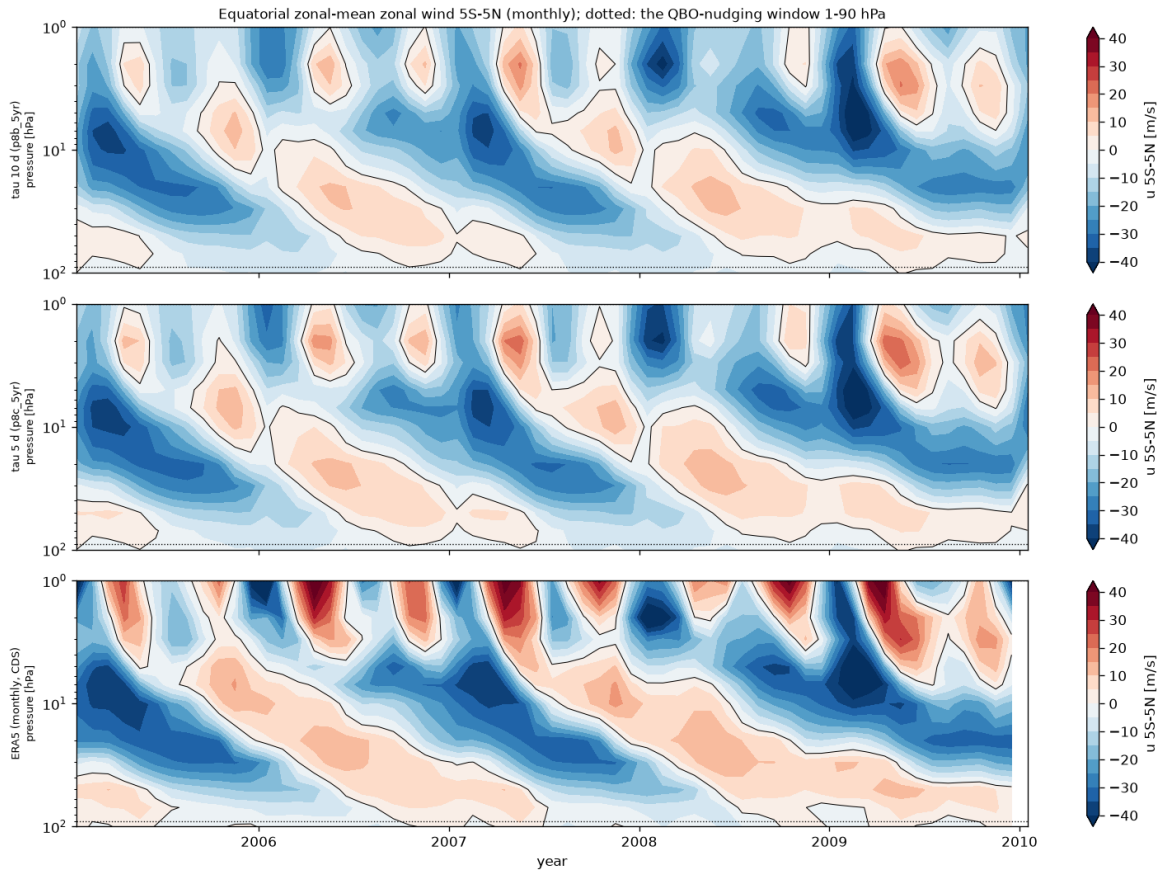


Figure 15. Equatorial wind 2005 to 2009 at tau 10 days (top), tau 5 days (middle) and ERA5 (bottom). The westerly phases at 5 to 20 hPa are broader and about 5 m/s stronger and the semiannual westerlies at 1 to 3 hPa more distinct; the picture is otherwise the same.

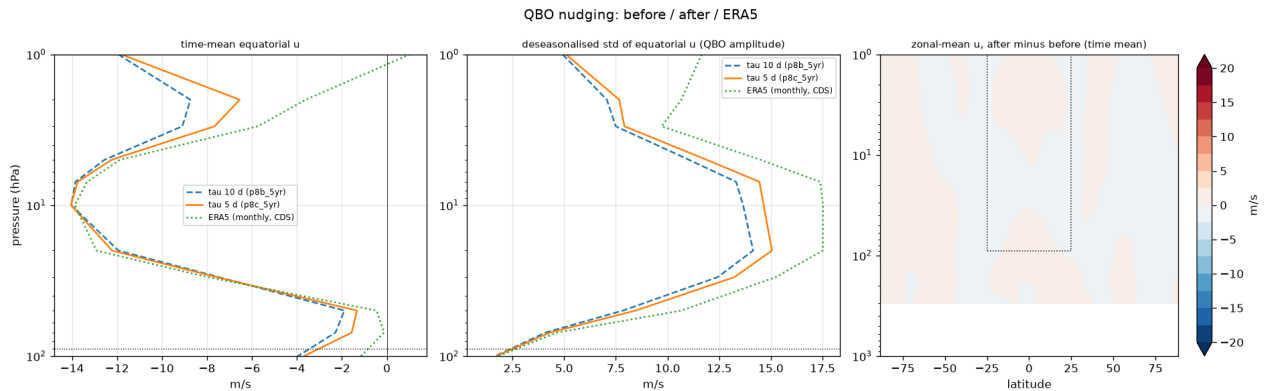


Figure 16. Left: the time-mean equatorial wind is unchanged. Middle: the QBO amplitude rises at every level by 6 to 9 percent. Right: the change in the time-mean zonal-mean wind is below 2 m/s everywhere.

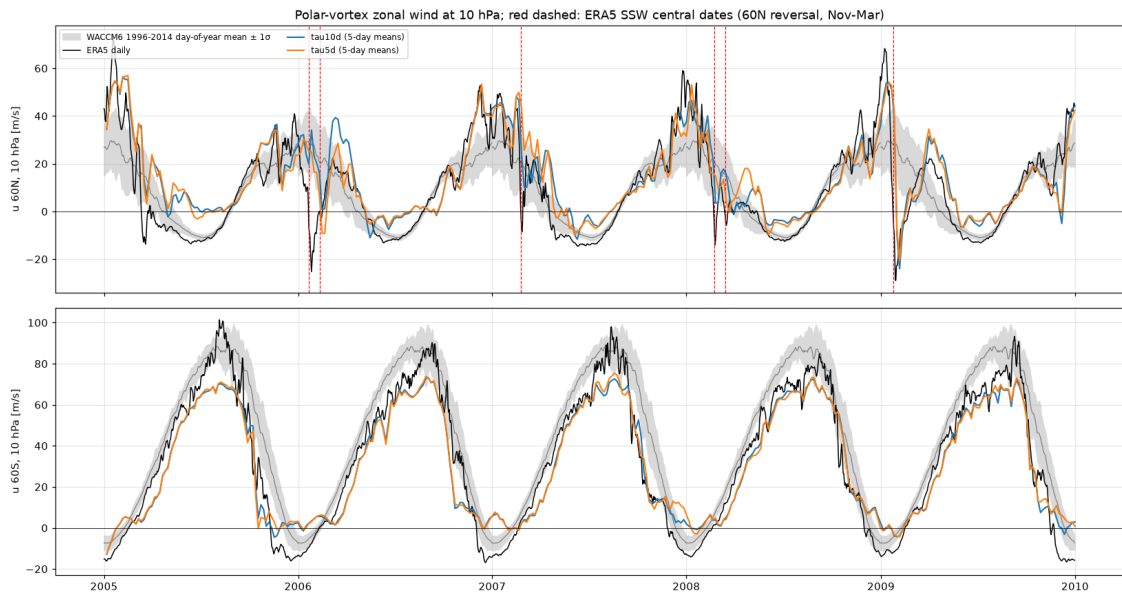


Figure 17. Polar-vortex wind at 10 hPa, tau 10 (blue) and tau 5 (orange) days. The one visible difference is a reversal in February 2006, a few days after ERA5's warming, which the tau 10 day chain did not produce; a single event far outside the window, read as internal variability.

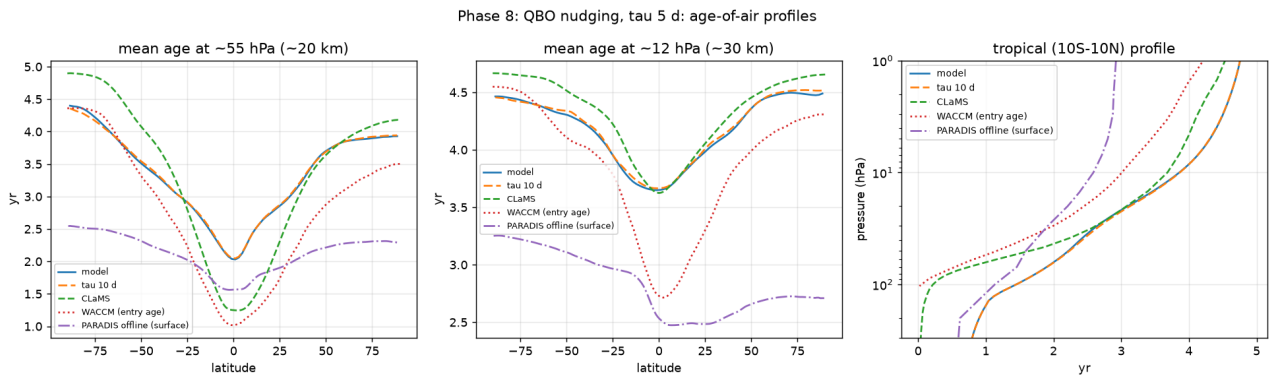


Figure 18. Age of air at tau 5 days (blue) against tau 10 days (orange dashed): identical to a hundredth of a year.

Halving tau closed about a quarter of the remaining amplitude gap at every level, as the two-spring estimate predicts, and nothing else in the model moved. The sweep was then completed with 2 days (runs p8d_*) and 1 day (p8e_*), same protocol.

The whole sweep

tau	std 10 / 20 / 30 / 50 hPa [m/s]	percent of ERA5 at 20 hPa (two-spring prediction)	RMS vs ERA5, 10-70 hPa	RMS, 1-7 hPa	SAO 1 / 2 / 3 hPa	u RMSE 100-1 hPa
10 d	13.7 / 14.1 / 12.4 / 7.8	81 (80)	4.0	12.5	9.5 / 15.8 / 13.8	4.9 m/s
5 d	14.6 / 15.1 / 13.2 / 8.5	86 (89)	3.1	10.9	12.2 / 17.3 / 13.9	4.8
2 d	15.4 / 15.8 / 13.9 / 9.0	90 (95)	2.5	8.9	14.9 / 18.1 / 13.7	4.6
1 d	15.7 / 16.1 / 14.2 / 9.3	92 (98)	2.3	7.7	17.2 / 18.2 / 13.7	4.4
ERA5	17.5 / 17.5 / 15.2 / 10.7	100	-	-	30.7 / 20.7 / 15.6	-

Unchanged across the sweep: T RMSE 100-1 hPa 6.3 to 6.2 K; jets 31 / 64 to 30 / 64 m/s; Brewer-Dobson 70 hPa 7.7 to 8.2 $\times 10^9$ kg/s; age of air at 20 km 2.16 to 2.13 yr in the tropics, contrast 1.64 to 1.65; tracer conservation and throughput identical. Time-mean wind change against tau 10 days: 0.7 / 1.4 / 1.8 m/s inside the window, at most 0.5 outside it, 0.0 in the troposphere. Vortex reversals: a February 2006 reversal (ERA5 warming 11 February) appears at 5, 2 and 1 days and never at 10; the 2009 date wanders and the March 2008 reversal is missing at 1 day. Plausibly a Holton-Tan-type re-timing of wave events, but three chains cannot separate it from internal variability.

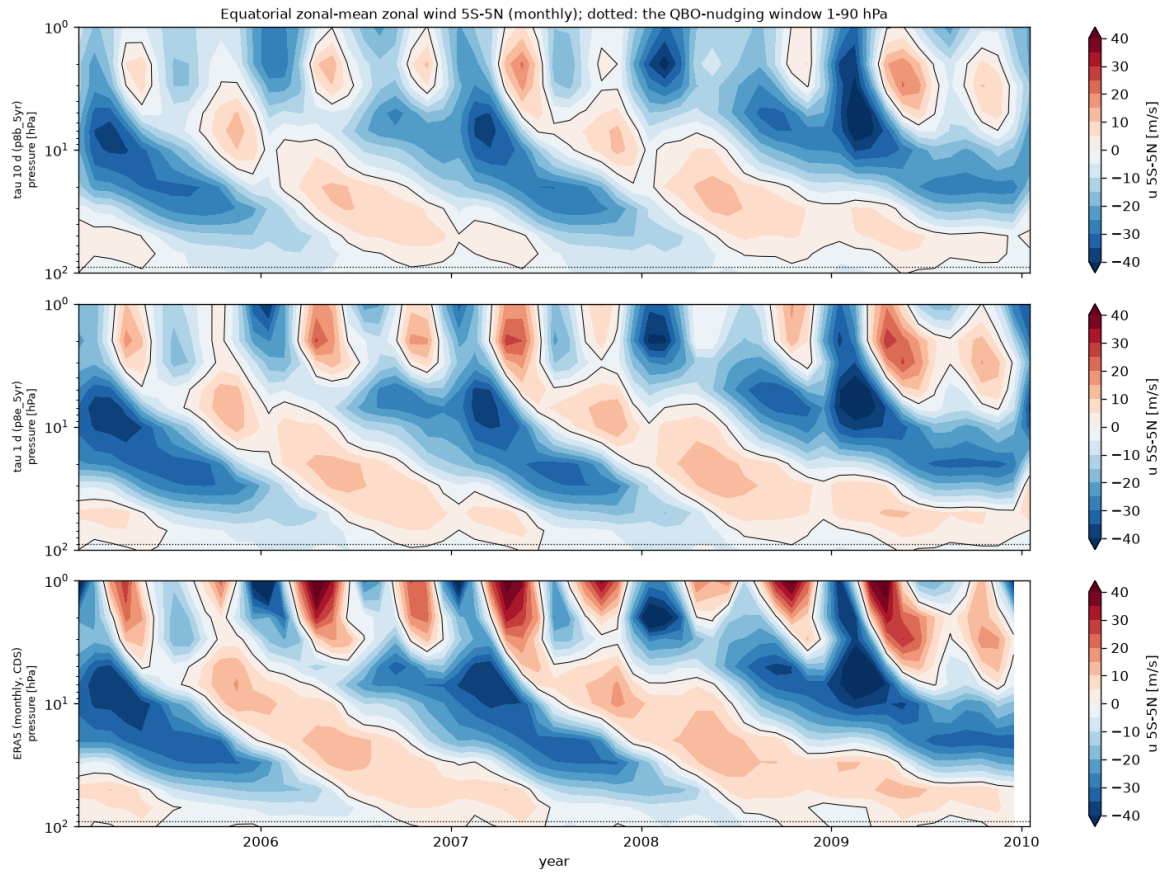


Figure 19. Equatorial wind 2005 to 2009 at tau 10 days (top), tau 1 day (middle) and ERA5 (bottom). At 1 day the westerly phases have ERA5's width and strength to within a few m/s, the onsets at 5 to 10 hPa are in place, and the semiannual oscillation at 1 to 3 hPa is distinct.

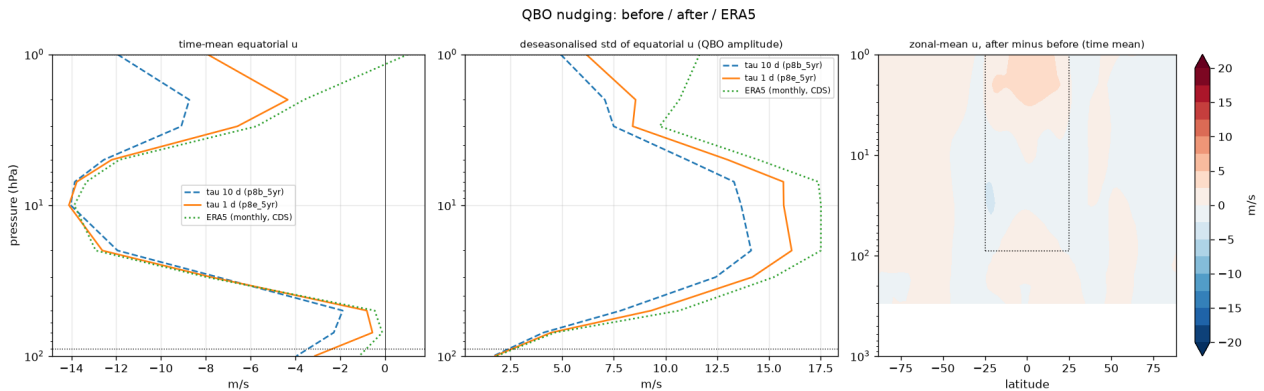


Figure 20. Left: the time-mean equatorial wind at tau 1 day (orange) lies on ERA5 (green) from 40 hPa to 1.5 hPa. Middle: the QBO amplitude at 90 to 93 percent of ERA5 at every level. Right: the change against tau 10 days stays inside the window.

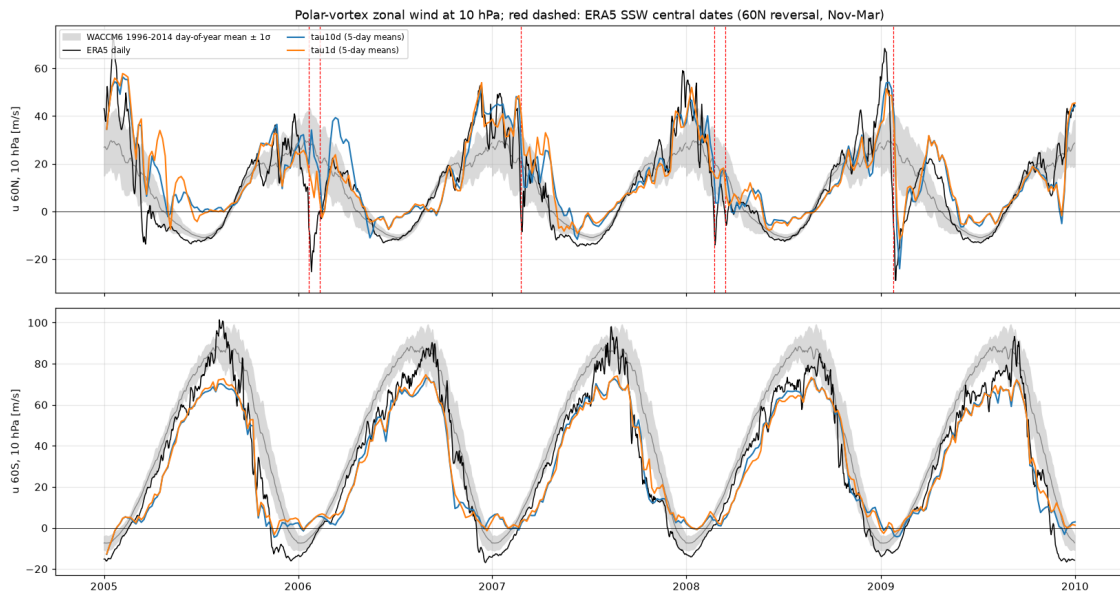


Figure 21. Polar-vortex wind at 10 hPa at tau 10 (blue) and 1 day (orange). The February 2006 reversal, and otherwise two curves on top of each other.

Reading the sweep

The amplitude follows the two-spring law down to 2 days and then flattens: 80, 86, 90, 92 percent against the predicted 80, 89, 95, 98. From 2 to 1 day the gain is 2 percent, so the last 8 percent is not the relaxation any more but the target itself. Above the QBO layer the gain does not flatten (12.5 to 7.7 m/s), because the semiannual signal is fast enough for tau to still matter. Nothing outside the window moves at any tau, and the global stratospheric wind error improves monotonically, all of it in the tropics.

Decision (KEY_DECISIONS 27): tau 1 day is the default. It is the value at which the relaxation stops being the limit; the model still sets its own temperature, since the thermal-wind adjustment takes about a day and the wind therefore does not run ahead of it as it would at 6 hours; and it sits in the range specified-dynamics models use (SD-WACCM: 50 hours). The daily-versus-monthly target question is settled the other way: the term already interpolates the monthly means to every 12-minute step, a daily target would impose sub-monthly changes that are not QBO, and what the target needs is the right shape, which is section 8.

8. The target's shape: mean-preserving interpolation

The target is ERA5's monthly-mean zonal wind, and the amplitude is scored against those same monthly means. The term interpolates linearly between month-centre values. A straight line between the centres of two neighbouring months cuts the corner of every peak, so the interpolant's own monthly means are smaller than the values it was drawn through: for a sinusoid of period P months by the factor $(6 + 2 \cos(2 \pi / P)) / 8$, which is 0.6 percent for the QBO's 28-month period and 12.5 percent for the 6-month semiannual oscillation. The AMIP boundary-condition method (Taylor et al. 2000) fixes this by solving for month-centre node values whose piecewise-linear interpolant reproduces the monthly means exactly; with equal months that is the tridiagonal system $(v[k-1] + 6 v[k] + v[k+1]) / 8 = m[k]$, ends clamped. Implemented as `qbo.mean_preserving` (default on, `mean_preserving_nodes` in `qbo_nudging.py`, unit-tested to $1e-3$ m/s); runs `p8f_2005` to `p8f_2009` at `tau 1 day` against the `tau 1 day` chain with plain interpolation, `p8e_5yr`. The cadence stays monthly; only the shape changes.

Results, 2005 to 2009

2005-2009	deseasonalised std 10 / 20 / 30 / 50 hPa [m/s]	mean u 20 / 30 hPa [m/s]	RMS vs ERA5, eq. monthly u 10-70 hPa [m/s]	RMS vs ERA5, 1-7 hPa [m/s]
tau 1 d, linear target (p8e_5yr)	15.7 / 16.1 / 14.2 / 9.3	-12.6 / -7.8	2.3	7.7
tau 1 d, mean-preserving target (p8f_5yr)	15.9 / 16.2 / 14.3 / 9.4	-12.6 / -7.8	2.2	7.1
ERA5 (monthly, CDS)	17.5 / 17.5 / 15.2 / 10.7	-12.9 / -8.0	0.0	0.0

Stratospheric climatology vs ERA5, 100-1 hPa, annual: meanpres: T RMSE 6.2 K, u RMSE 4.4 m/s, jets 14 (ref 10) / 32 (ref 33).

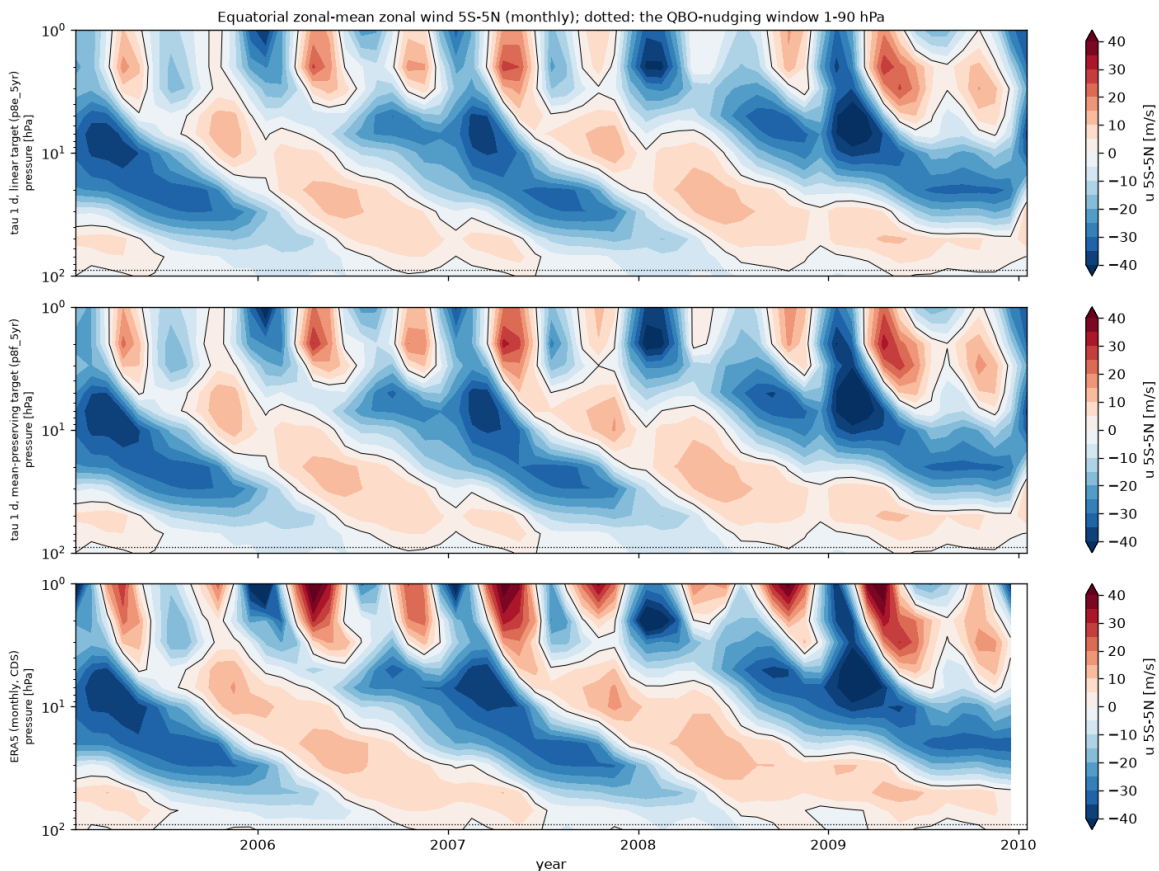


Figure 22. Equatorial wind 2005 to 2009: tau 1 day with plain interpolation (top), with mean-preserving nodes (middle), ERA5 (bottom).

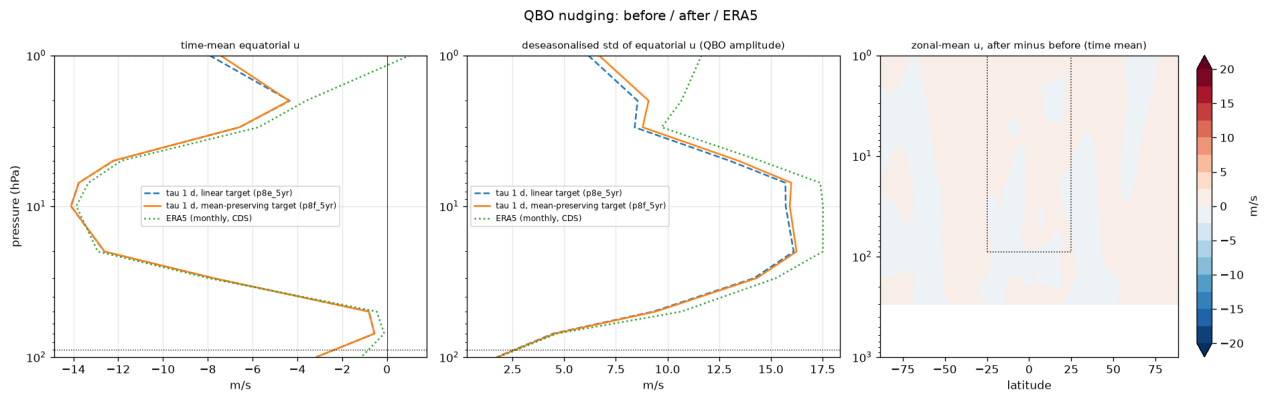


Figure 23. Time-mean equatorial wind, QBO amplitude and the change in the time-mean zonal-mean wind, mean-preserving minus plain.

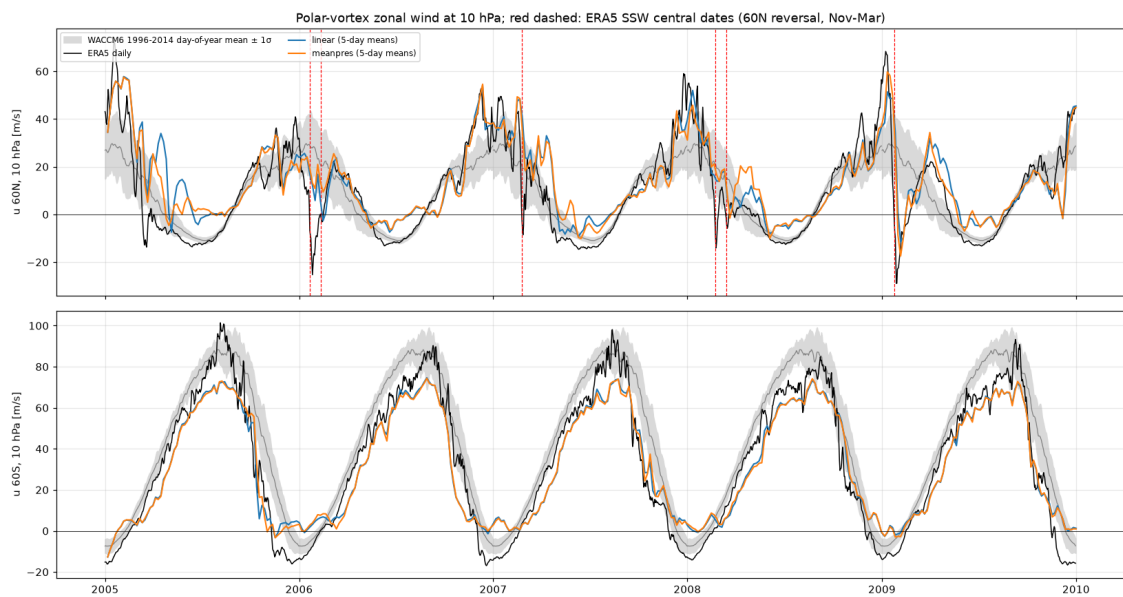


Figure 24. Polar-vortex wind at 10 hPa, both versions.

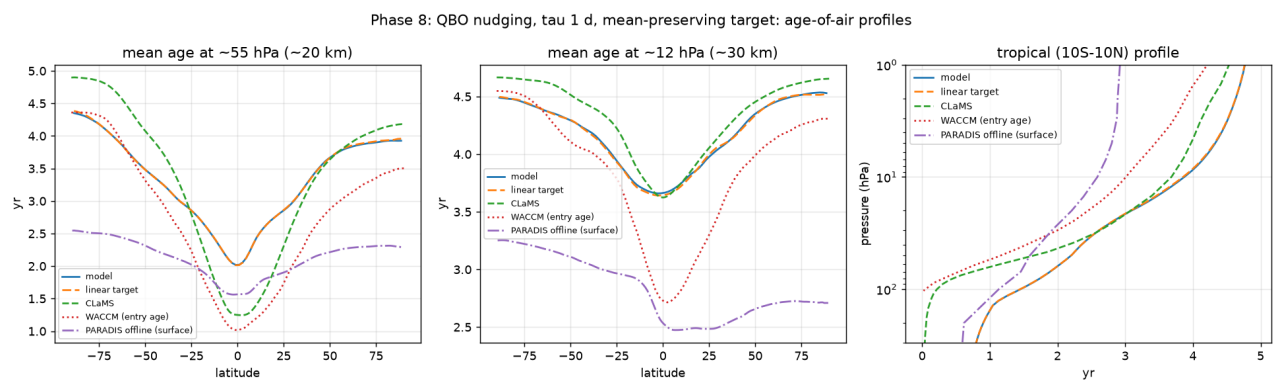


Figure 25. Age of air, mean-preserving (blue) against plain interpolation (orange dashed).

Reading

Exactly the filter prediction. The QBO amplitude gains 0.6 percent (16.1 to 16.2 m/s at 20 hPa, 92 to 93 percent of ERA5) and the semiannual oscillation gains 12 to 14 percent: at 2 and 3 hPa it now matches ERA5 to 0.1 m/s (20.7 / 15.5 against 20.7 / 15.6); at 1 hPa it stays at 19 against 31 because the nudging weight is zero there by construction. The error above the QBO layer falls from 7.7 to 7.1 m/s, inside it from 2.3 to 2.2. The remaining 7 percent of QBO amplitude is the model, not the target: the model's own easterly pull over one day, plus what the 5-day output means and the monthly scoring smooth away. Nothing else moved: the time-mean wind changes by 0.1 m/s inside the window and 0.3 outside, climatology, jets, Brewer-Dobson flux, age of air (2.13 years in the

tropics at 20 km in both) and tracers are identical.

The vortex reversals shuffled again: the February 2006 event that appeared at tau 5, 2 and 1 days with the linear target is absent here and the March 2008 one is back. Two chains that differ only by a 0.6 percent change of the tropical target giving different reversal dates settles the question left open in section 7: the reversal timing is internal variability of the wave events, not a response to the nudging.

Decision (KEY_DECISIONS 27): the mean-preserving target is the default, together with tau 1 day and the 1 hPa window top. The linear-target chains p8b to p8e stay on disk as before-states. This is the QBO configuration Phases 9 and 10 inherit.

9. Reading and what comes next

- **The model now has ERA5's QBO, in phase, at 80 percent amplitude, and with the window top at 1 hPa also ERA5's semiannual oscillation above it.** The equatorial wind error against ERA5 fell from 16 m/s (Phase 6) to 4 in the QBO layer and, with the 1 hPa top, from 23 to 12 above it. The global stratospheric wind error, 5.6 m/s in Phase 6, is 4.9 with the 1 hPa top: the nudging no longer costs anything in the climatology.
- **The rest of the stratosphere is untouched.** Polar-night jets, sudden warmings, polar-cap temperatures, the Brewer-Dobson flux and the tracer budgets are the Phase 6 ones within noise, in both versions. The nudging does what it says: the tropical mean flow and nothing else.
- **Age of air moves a little, in the right direction, and not much.** Tropical age at 20 km 2.29 to 2.16 years; contrast to the extratropics 1.56 to 1.63 (CLaMS 2.79). A five-year mean averages over two QBO cycles; the QBO's value for aerosol is in phase-dependent statements (residence time in the easterly against the westerly phase, subtropical leakage), which the model can now make. The remaining 0.9 year tropical excess is the Phase 4 diagnosis unchanged (issue 25).
- **The window top was not what limits the westerly phases; tau was, and then the target's shape.** Raising the top to 1 hPa left the QBO layer exactly as it was. The tau sweep 10, 5, 2, 1 days gave 80, 86, 90, 92 percent of ERA5's amplitude with nothing else moving, following the two-spring estimate (model restoring time about 40 days) down to 2 days and flattening below, where the linear interpolation cutting the peaks of the monthly target becomes the limit. Above the QBO layer the gain continued to 1 day and the semiannual oscillation at 2 hPa reached ERA5. No artefact appeared at the window edges from the stiff wind spring against the 15-day Polvani-Kushner relaxation; the age of air moved by 0.03 years over the whole sweep.
- **Cadence versus shape.** The term already interpolates the monthly target to every 12-minute step, so a daily target would not change the cadence the model sees; it would only add sub-monthly zonal-mean wind changes that are not QBO. What the target needed was the right shape, mean-preserving nodes, which is a property of how the monthly means are interpolated, not of their cadence.
- **Decisions.** Window top 1 hPa (KEY_DECISIONS 26) and tau 1 day with the mean-preserving target (KEY_DECISIONS 27) are the defaults. The 4 hPa chain and the tau 10, 5, 2 day chains stay on disk as before-states. Open: whether the February 2006 vortex reversal that appears at every short tau is a Holton-Tan-type response (DEFERRED.md).
- **Cost.** The zonal-mean reduction adds about 1 ms to a 7 ms step, minus 13 percent stepping throughput, the same for both window tops; a cheaper reduction on the dycore grid is issue 44. A five-year chain takes 70 to 80 minutes on one GPU.

10. Where things are

Runs under runs/p8_* (window top 4 hPa), runs/p8b_* (1 hPa), runs/p8c_*, p8d_*, p8e_* (1 hPa, tau 5 / 2 / 1 days) and runs/p8f_* (tau 1 day, mean-preserving target); record docs/outputs/08_qbo/output.md with the figures beside it (5yr/, 1yr/, 1hpa_top/, tau5/, tau2/, tau1/, interp/); the term jcm_strat/qbo_nudging.py with tests/test_qbo_nudging.py; configuration jcm_strat/config/physics/strat_pk_qbo.yaml and experiment p8_qbo; the comparison script scripts/qbo_compare.py; the ERA5 target fetch scripts/fetch_era5_strat_ref.py (cache/era5_ref/). Decisions 23, 24 and 26 in KEY_DECISIONS.md; open items: issue 44 (cheaper zonal mean), issue 6 (close on merge).